

Computer Networks Lab 04

Course: Computer Networks (CL3001) | Semester: Fall 2026

Course Coordinator: Ms. Ramsha Iqbal

Learning Objectives

- Understand the working of HTTP and HTTPS protocols and their differences in terms of security, ports and layers.
- Analyze HTTP status codes (Client Errors 4xx and Server Errors 5xx) and their significance.
- Implement and simulate HTTP/HTTPS services in Cisco Packet Tracer and observe headers using Wireshark simulation mode.
- Understand DNS hierarchy, record types (A, CNAME, NS, SOA) and configure DNS server for domain name resolution.
- Understand basic router functionality, routing table components and perform simple IP assignment on router interfaces using GUI and CLI.

Introduction to Routers

A router is a networking device that connects two or more different networks together and forwards data packets between them. It determines the best path for data to travel from source to destination. In simple words, a router is like a traffic policeman for your network that directs internet traffic.

How Does a Router Work?

1. A router operates at Layer 3 (Network Layer) of the OSI model.
2. When a data packet arrives, the router reads its destination IP address.
3. It checks its Routing Table to find the best path to reach that destination.
4. It then forwards the packet to the next router (next hop) or directly to the destination network.
5. This process continues until the packet reaches its final destination.

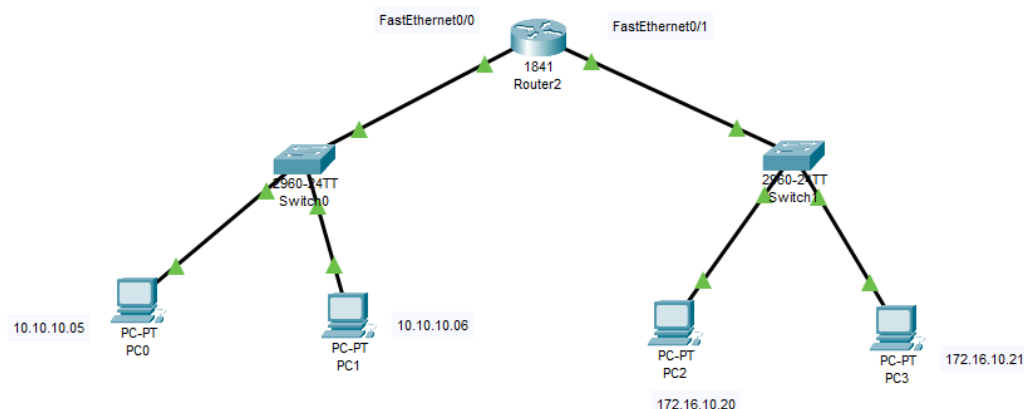
Every router has its own CPU, RAM, operating system (like Cisco IOS), and interfaces (ports) to connect cables.

Key Functions of a Router:

- **Path Determination:** Finds the best path using routing protocols like RIP, OSPF, EIGRP.
- **Packet Forwarding:** Moves packets from one interface to another.
- **Traffic Filtering:** Can block or allow traffic based on rules (ACL).
- **Network Segmentation:** Divides a large network into smaller sub-networks.

Implementation for Router

Design the given topology shown below. Assign IP address to PCs mentioned in the diagram using static.



```

Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 10.10.10.1 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#
Router(config-if)#exit
  
```

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 172.16.10.1 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

Router(config-if)#exit
```

HTTP/HTTPS

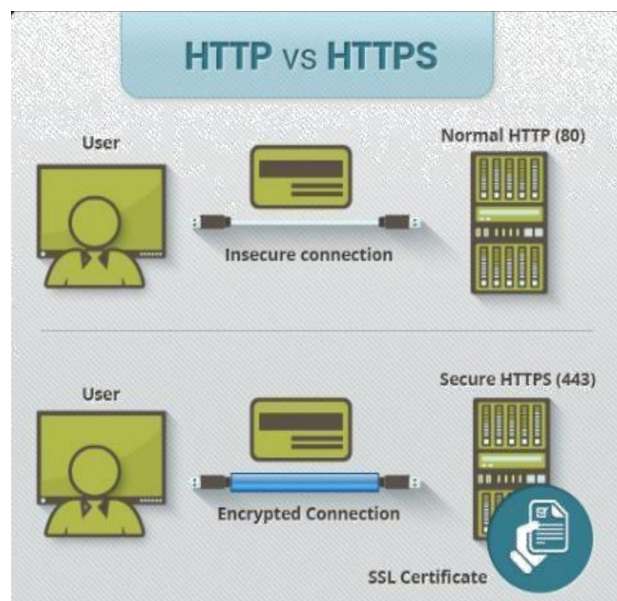
1. Hypertext Transfer Protocol (HTTP)

Hypertext Transfer Protocol (HTTP) is a protocol used in networking. When you type any web address in your web browser, your browser acts as a client and the computer having the requested information acts as a server. When client requests for any information from the server, it uses HTTP protocol to do so. The server responds back to the client after the request completes. The response comes in the form of web page which you see just after typing the web address and press "Enter". HTTP uses Port 80 for communication.

2. Hypertext Transfer Protocol Secure (HTTPS)

Hypertext Transfer Protocol Secure (HTTPS) is a combination of two different protocols. It is more secure way to access the web. It is combination of Hypertext Transfer Protocol (HTTPS) and SSL/TLS protocol. It is more secure way to sending request to server from a client, also the communication is purely encrypted which means no one can know what you are looking for. This kind of communication is used for accessing those websites where security is required. Banking websites, payment gateway, emails (Gmail offers HTTPS by default in Chrome browser), and corporate sector websites are some great examples where HTTPS protocols are used. HTTPS uses Port 443 for communication.

For HTTPS connection, public key trusted and signed certificate is required for the server. These certificates come either free or it costs few dollars depends on the signing authority. There is one other method for distributing certificates. Site admin creates certificates and loads in the browser of users. Now when user requests information to the web server, his identity can be verified easily.



3. Difference between HTTP & HTTPS

Below given are some major differences between HTTP & HTTPS:

HTTP	HTTPS
URL begins with "http://"	URL begins with "https://"
It uses port 80 for communication.	It uses port 443 for communication.
Unsecured.	Secured.
Operates at Application Layer.	Operates at Transport Layer.
No encryption.	Encryption is present.
No certificates required.	Certificates are required.

4. Client Error

The 4xx class of status code is intended for cases in which the client seems to have erred. Except when responding to a HEAD request, the server should include an entity containing an explanation of the error situation, and whether it is a temporary or permanent condition. These status codes are applicable to any request method. User agents should display any included entity to the user.

400 Bad Request

The server cannot or will not process the request due to something that is perceived to be a client error (e.g., malformed request syntax, invalid request message framing, or deceptive request routing).

401 Unauthorized (RFC 7235)

Similar to 403 Forbidden, but specifically for use when authentication is required and has failed or has not yet been provided. The response must include a WWW-Authenticate header field containing a challenge applicable to the requested resource. See Basic access authentication and Digest access authentication.

403 Forbidden

The request was a valid request, but the server is refusing to respond to it. Unlike a 401 unauthorized response, authenticating will make no difference.

404 Not Found

The requested resource could not be found but may be available again in the future. Subsequent requests by the client are permissible.

408 Request Timeout

The server timed out waiting for the request. According to HTTP specifications: "The client did not produce a request within the time that the server was prepared to wait. The client may repeat the request without modifications at any later time."

5. Server Error

The server failed to fulfill an apparently valid request.

Response status codes beginning with the digit "5" indicate cases in which the server is aware that it has encountered an error or is otherwise incapable of performing the request. Except when responding to a HEAD request, the server should include an entity containing an explanation of the error situation, and indicate whether it is a temporary or permanent condition. Likewise, user agents should display any included entity to the user. These response codes are applicable to any request method.

500 Internal Server Error

A generic error message, given when an unexpected condition was encountered and no more specific message is suitable.

501 Not Implemented

The server either does not recognize the request method, or it lacks the ability to fulfil the request. Usually this implies

future availability (e.g., a new feature of a web-service API).

502 Bad Gateway

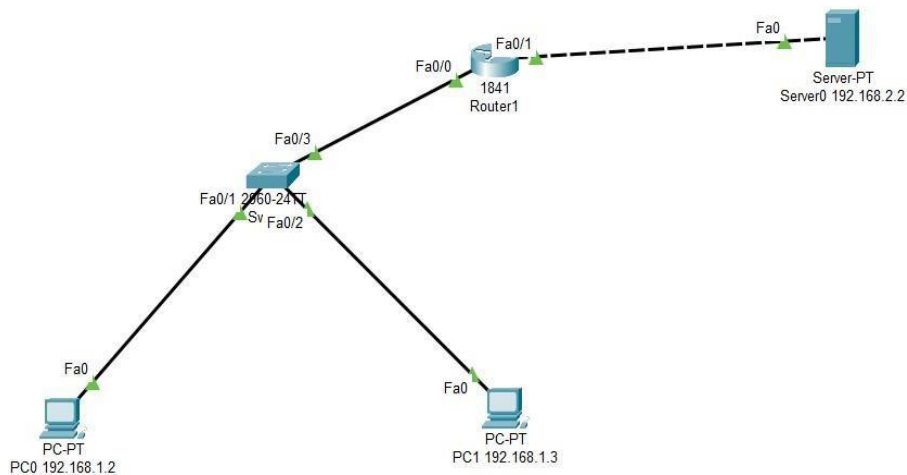
The server was acting as a gateway or proxy and received an invalid response from the upstream server.

503 Service Unavailable

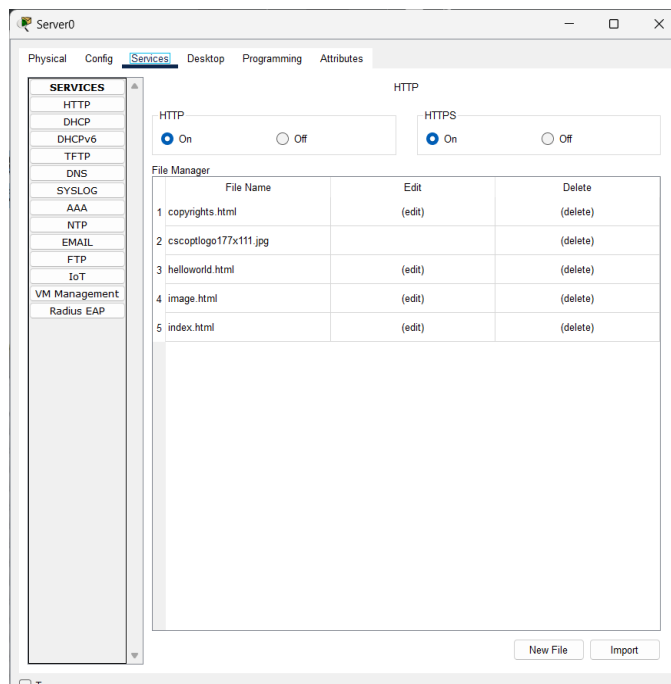
The server is currently unavailable (because it is overloaded or down for maintenance). Generally, this is a temporary state.

6. Implementation for HTTP

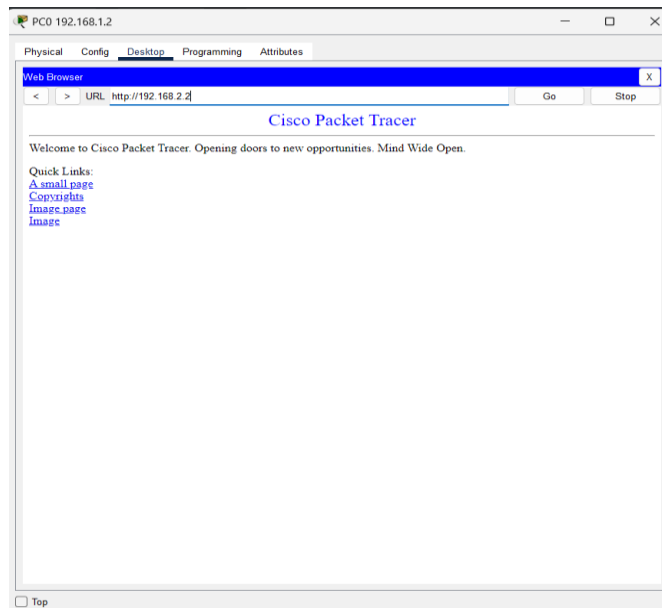
Design the given topology shown below. Assign IP address to PC using static through as done in pervious lab.



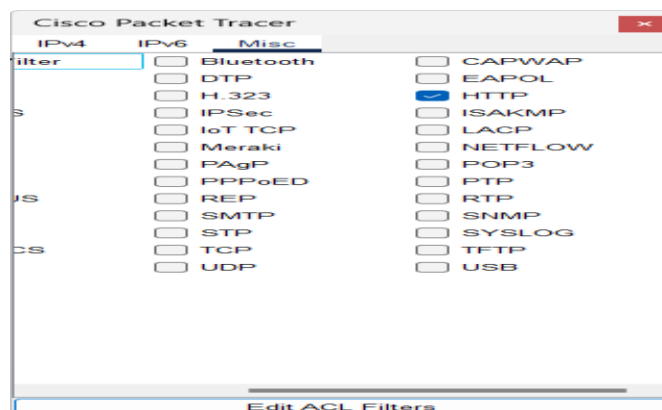
The above topology configured as "one server room", "one IT room" and "Lab#01 environment having two systems". On our server we have enabled web services as well as DNS services. Click on the web server, go to services tab—HTTP. Here you can see HTTP & HTTPS services are on.



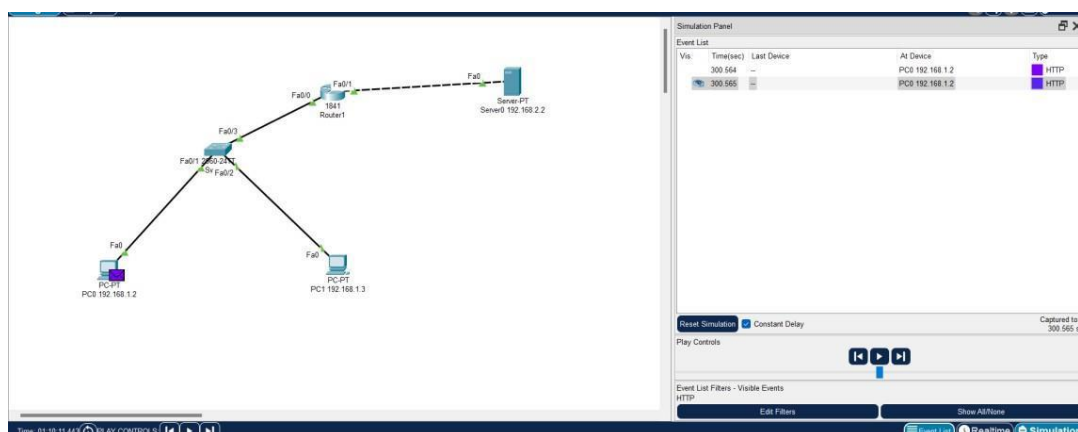
Now click on PC0 and go to Desktop -> Web Browser. Now type web-server IP which you have assign.

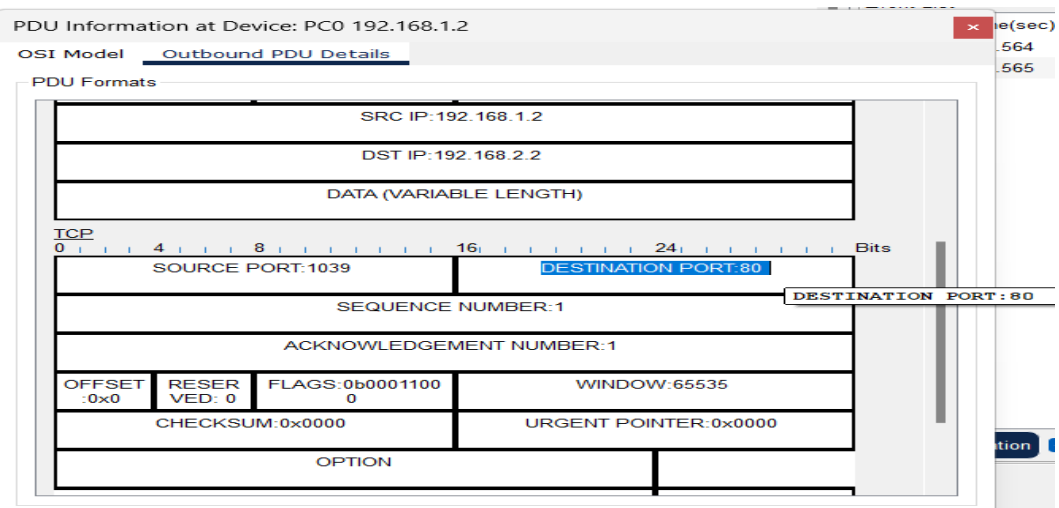


To note the http header format information, go to simulation mode, edit filters and click on http check box then click on capture/forward button.



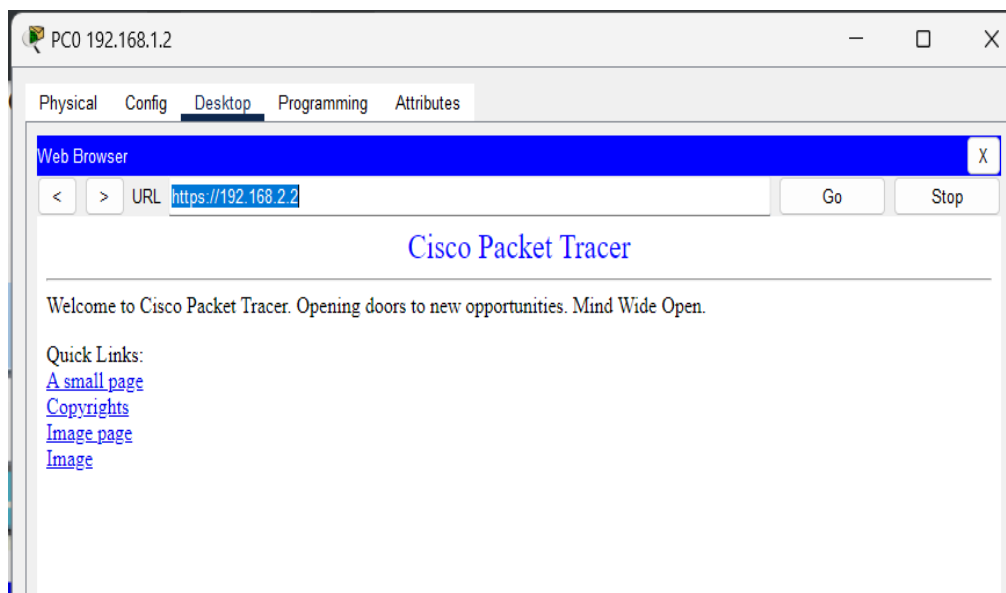
Now click on the http packet, you can note that the destination port is 80.



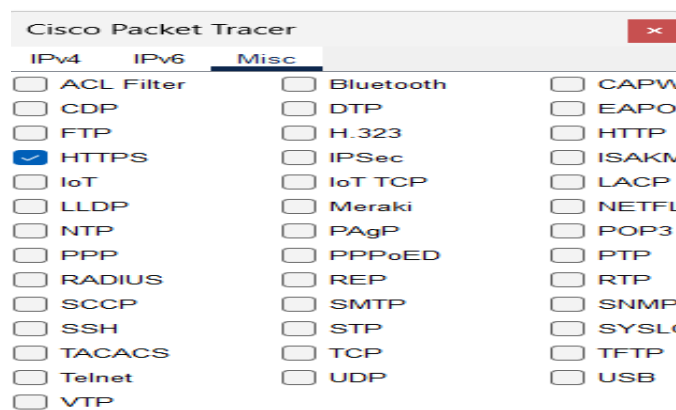


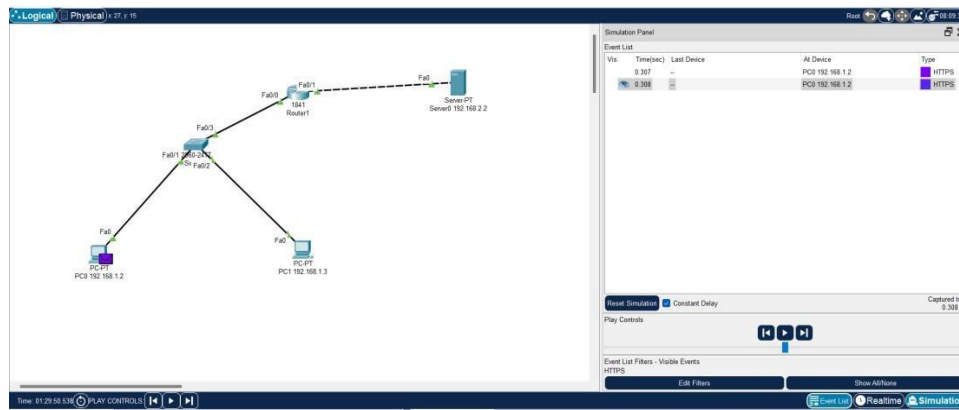
7. Implementation for HTTPS

Now click on PC and go to Desktop>Web Browser. Now type web-server IP 192.168.2.2

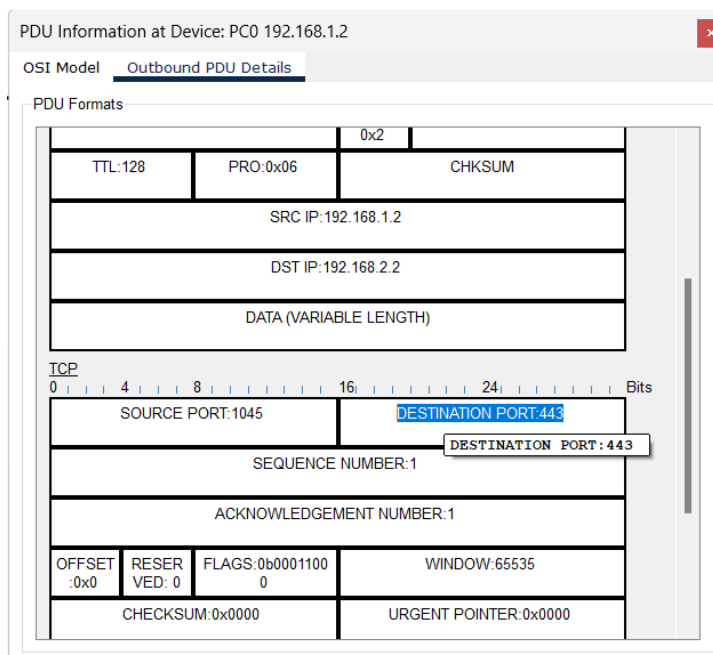


Now to note the https header format information go to simulation mode----- > edit filters and click on https check box then click on capture/forward button.





Now click on the https packet, you can note that the destination port is 443.



Now scroll the Outbound PDU Details, you can see the https PDU.

DNS in Cisco Packet Tracer

1. Introduction to DNS

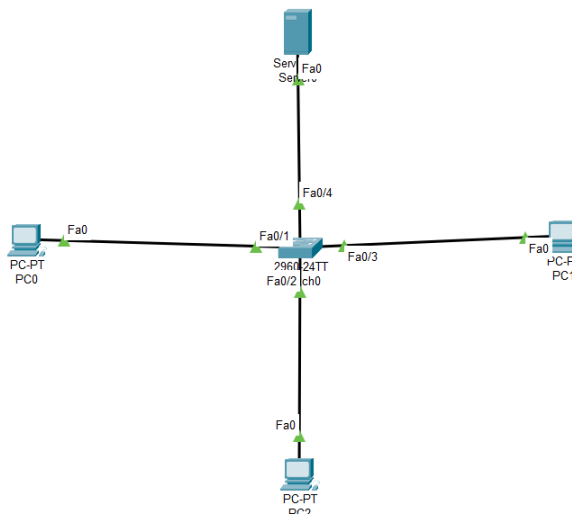
The Domain Name System (DNS) is a hierarchical and distributed naming system for computers, services, and other resources in the Internet. It translates readily memorized domain names to the numerical IP addresses needed for locating and identifying computer services.

Some common DNS record types are:

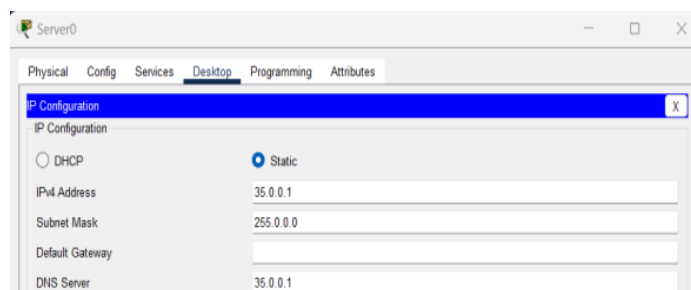
- a) A Record:** An A record is actually an address record, which means it maps a fully qualified domain name (FQDN) to an IP address. For example, google.com to 74.125.224.147.
- b) CNAME Record:** Canonical name records, or CNAME records, are often called alias records because they map an alias to the canonical name.
- c) NS Record:** An NS record identifies which DNS server is authoritative for a particular zone.
- d) SOA Record:** The SOA or Start of Authority record for a domain stores information about the name of the server that supplies the data for the zone.

2. Implementation of DNS

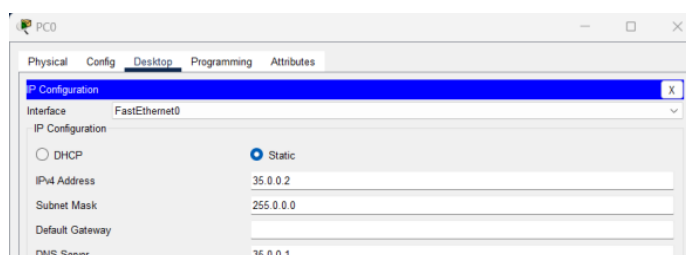
Consider the following topology:

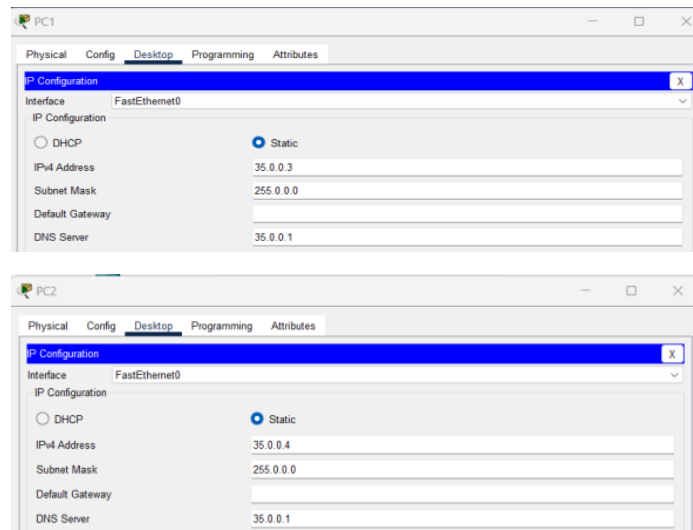


Assigning IP to DNS server:



Assigning IP to PCs:

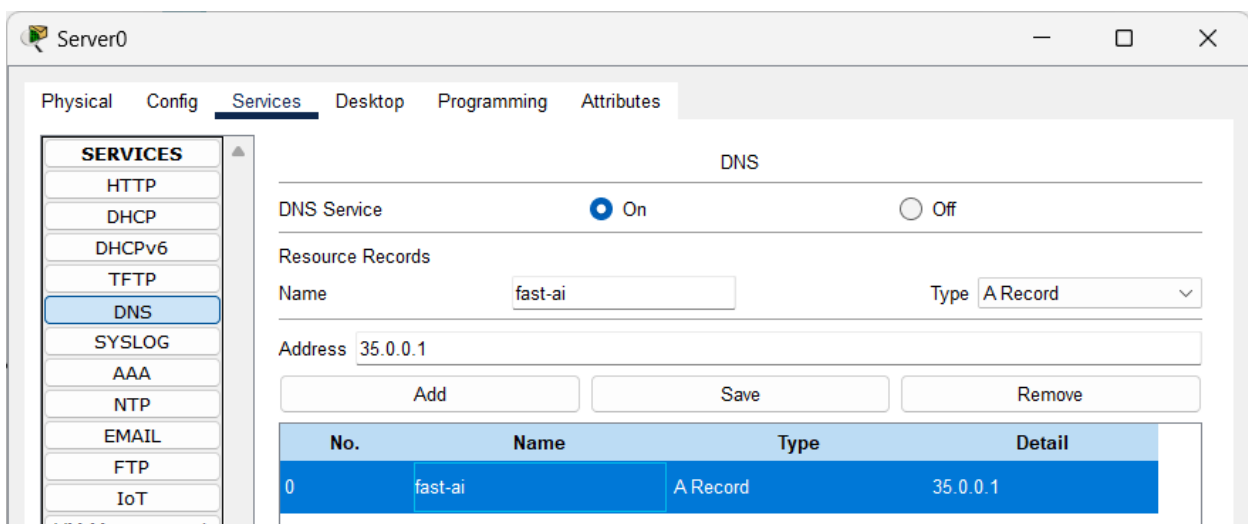




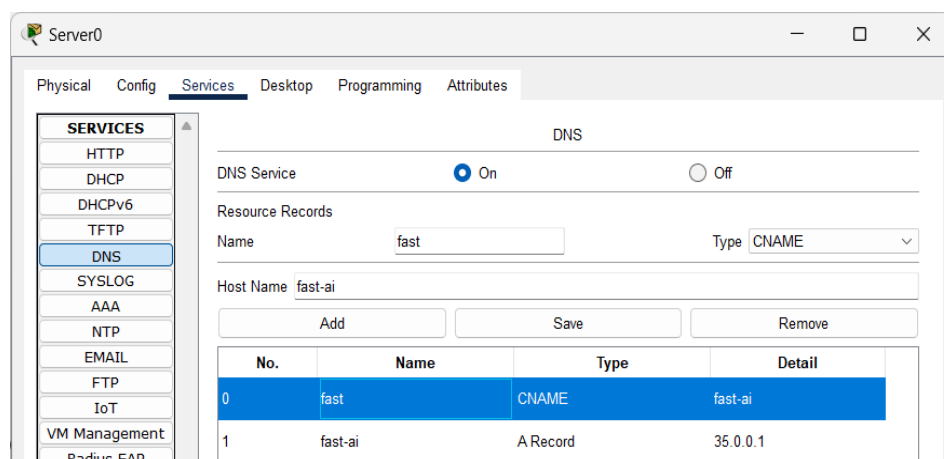
1. DNS Configuration & Simulation

Now using the DNS service on DNS Server. Go to server services DNS.

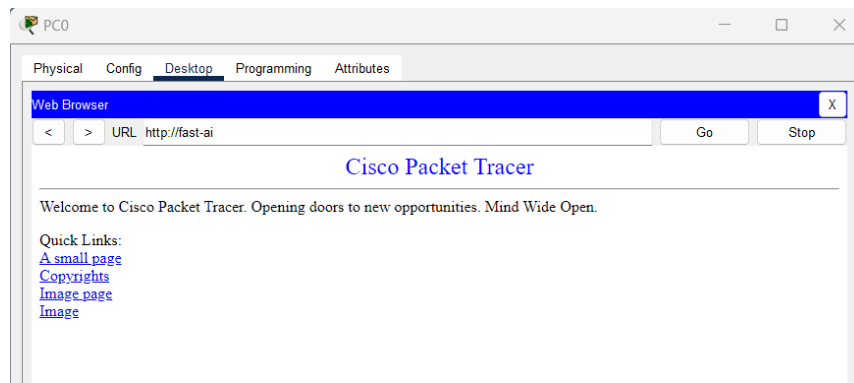
First, we add A record. We assign the web server IP against our Domain name. Now click on Add.



Now add CNAME record.



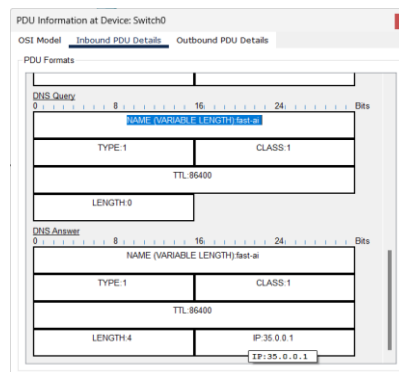
Now go to any PC → Desktop → web browser → type **“fast-ai”** and see how DNS works.



Start simulation.

Vis.	Time(sec)	Last Device	At Device	Type	Info
	0.000	--	PC4	DNS	
	0.001	PC4	Switch1	DNS	
	0.002	Switch1	DNS Ser...	DNS	
	0.003	DNS Server	Switch1	DNS	
	0.004	--	PC4	TCP	
	0.004	Switch1	PC4	DNS	
	0.004	--	PC4	TCP	
	0.005	PC4	Switch1	TCP	
	0.006	Switch1	Web Ser...	TCP	

Click on DNS packet. See how DNS server resolved the name.



Shows OSI layers involved in transmission.

